Vivekanand Education Society's
Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai, Approved by AICTE & Recognised by Govt. of Maharashtra)

NAAC accredited with 'A' grade

PROJECT REPORT ON

CogniSense : Multimodal Digital Biomarkers for Pre-clinical Dementia Screening

SUBMITTED IN FULFILLMENT OF THE REQUIREMENT FOR SEMESTER VI OF

T.E.(Information Technology)

SUBMITTED BY

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V.E.S. INSTITUTE OF TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY

2025-26

Vivekanand Education Society's
Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai, Approved by AICTE & Recognised by Govt. of Maharashtra)

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Certificate

This is to certify that project entitled

"CogniSense : Multimodal Digital Biomarkers for Pre-clinical Dementia Screening"

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Declaration

I declare that this written submission represents my ideas in my own words. Wherever the ideas or words of others have been included, I have adequately cited and referenced the original sources. I further declare that I have adhered to all principles of academic honesty, integrity, and ethical conduct in the preparation of this work.

I affirm that I have neither misrepresented nor fabricated nor falsified any idea, data, fact, or source in my submission. I have ensured that all information presented is accurate to the best of my knowledge and obtained through ethical means.

I understand that any violation of the above principles may lead to disciplinary action by the Institute and may also invite penal action from the concerned sources if proper citation or permission has not been provided where required.

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Abstract

Alzheimer's Disease and related dementias represent a growing global health crisis, with India facing a significant diagnostic gap due to high costs and a lack of specialized neurologists. While **theoretical** models of neurodegeneration emphasize the critical need for early intervention, current clinical diagnoses, such as MRIs and PET scans, typically occur at the "Symptomatic Stage", by which point irreversible neuronal damage has already occurred.

Addressing limitations in current **literature**, this project proposes a smartphone-based "Digital Triage" system designed to detect pre-clinical signs of Alzheimer's 5–7 years before visible symptoms appear. Our **methodological** approach leverages Deep Learning and Data Warehouse (DWH) Mining to analyze a multimodal set of digital biomarkers. These biomarkers **include** acoustic speech patterns, oculomotor (eye) movements, and visuospatial drawing metrics. Unlike existing solutions that require specialized hardware, our framework utilizes standard smartphone sensors, making it an ideal tool for General Physicians and Family Doctors to use during routine geriatric checkups.

The inclusion of DWH Mining allows for longitudinal tracking of a patient's "Brain Velocity", identifying subtle rates of decline over time. This non-invasive approach aims to bridge the diagnostic gap, providing an affordable, language-agnostic screening tool for the Indian middle-class and rural populations.

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Chapter 1 : Introduction

1.1 Introduction to the Project

Alzheimer's Disease (AD) and related dementias represent a growing global health crisis, with over 55 million people currently living with dementia worldwide. In India alone, the Dementia India Report estimates approximately 8.8 million individuals are currently affected, with numbers expected to double every 20 years.

Historically, clinical diagnoses relied heavily on expensive and invasive tools such as Positron Emission Tomography (PET) scans, Cerebrospinal Fluid (CSF) analysis, and structural MRIs. These tools were deployed reactively, only after a patient or family member reported visible cognitive decline, such as memory lapses or personality changes. By that symptomatic stage, clinical studies confirmed that 40 to 60 percent of neurons in the hippocampus and entorhinal cortex had already been permanently lost. Consequently, the healthcare infrastructure in India faced a significant diagnostic gap due to a severe lack of specialized neurologists (0.07 per 100,000 population) and prohibitive diagnostic costs.

To address these challenges, CogniSense proposes a smartphone-based "Digital Triage" system designed to detect pre-clinical signs of Alzheimer's 5-7 years before visible symptoms appear. By leveraging Deep Learning and Data Warehouse (DWH) mining, the system analyzes a multimodal set of digital biomarkers—acoustic speech patterns, oculomotor (eye) movements, and visuospatial drawing metrics. Unlike existing solutions that require specialized hardware, our framework utilizes standard smartphone sensors, making it an ideal, accessible tool for General Physicians to use during routine geriatric checkups.

1.2 Objectives

The primary aim of this project is to develop CogniSense, a smartphone-based multimodal deep learning framework designed to detect pre-clinical cognitive markers of Alzheimer's Disease 5-7 years before symptomatic onset.
The specific objectives of the project are:

- To integrate acoustic speech analysis, oculomotor tracking, and visuospatial assessments into a unified cognitive risk score using a novel Cross-Attention Transformer fusion module.
- To establish a Data Warehouse (DWH) layer that enables longitudinal "Brain Velocity" tracking, quantifying the rate of cognitive decline across repeated patient assessments.
- To deploy a hardware-independent, language-agnostic "Step 0" digital screening tool for General Physicians using standard Android tablets.

1.3 Motivation

The fundamental tragedy of Alzheimer's Disease is not merely its prevalence, but the timing of its detection. The pathological cascade of AD begins 15 to 20 years before the first observable clinical symptom. This extended pre-clinical window represents a critical, largely untapped opportunity for intervention, yet existing healthcare infrastructure fails to exploit it. Existing diagnostic solutions are confined to specialist centers and require highly expensive clinical hardware, resulting in a systemic failure to detect cognitive decline early. Furthermore, current cognitive screening tools lack multimodal fusion, smartphone feasibility, and language-agnosticism.

CogniSense shifts the focus from reactive hospital-grade testing to proactive, community-based screening. By acting as a "Step 0 Digital Triage" tool, it improves upon current systems by moving the point of detection from the specialist's office to the local primary care clinic. This approach dramatically reduces the cost of screening and eliminates language barriers, ultimately benefiting patients, caregivers, and the broader healthcare system by providing early, manageable intervention opportunities.

1.4 Brief Introduction to Literature

Various studies have explored solutions and technologies relevant to this problem domain. Existing systems and research papers were analyzed to understand the methodologies and approaches used earlier. For instance, works by Fraser et al. and König et al. validated that automated speech analysis and natural language processing could detect pre-dementia patients with strong accuracy. Similarly, studies by Shafiq-Antonacci et al. established oculomotor dysfunction, specifically antisaccade error rates, as a highly sensitive pre-clinical biomarker.

It was observed that while several individual modality solutions existed, they had certain limitations in terms of efficiency, scalability, and accessibility. Oculomotor studies, for example, relied exclusively on expensive laboratory-grade hardware (costing ₹5-15 lakhs), precluding their use in primary healthcare settings. Furthermore, a systematic literature review by Petti et al. confirmed the validity of speech-based detection but explicitly identified the absence of multimodal fusion and cross-lingual validated systems as the field's primary limitations. This analysis provides a robust foundation for developing an improved, accessible, and multimodal solution like CogniSense that bridges these gaps.

1.5 Problem Definition

The current healthcare system faces severe challenges in the early diagnosis of Alzheimer's Disease, exacerbated by inefficiency, lack of objective automation at the primary care level, and limited specialist accessibility. Existing diagnostic methods—such as PET scans and full neuropsychological batteries—are highly costly, time-consuming, and deployed reactively only after a patient exhibits undeniable cognitive decline.

There is a critical need to address these issues by developing a more effective and reliable pre-clinical screening solution. The proposed project aims to overcome these limitations by introducing an affordable, hardware-independent, and longitudinally-aware screening tool that enables General Physicians to objectively assess cognitive decline during routine 5-minute checkups, preventing irreversible neuronal damage.

1.6 Scope of the Work

- **Functionalities:** The CogniSense system will capture three categories of digital biomarkers (speech acoustics, eye tracking, and a digitized Clock Drawing Test) via a standard Android tablet. It will process this data through independent deep learning pipelines (Wav2Vec 2.0, CNN+LSTM, Vision Transformers), fuse the outputs using a Cross-Attention Transformer, and generate an Explainable AI clinical report. A Data Warehouse will be implemented to store results and compute longitudinal "Brain Velocity".

- **Limitations and Constraints:** A positive flag from CogniSense does not constitute a final medical diagnosis; it acts strictly as a clinical decision support tool to recommend specialist referrals. Additionally, the computation of meaningful Brain Velocity requires a minimum of two longitudinal assessment records over time.

- **Target Users and Environment:** The system is designed specifically for General Physicians and Family Doctors to administer in primary healthcare centers, community clinics, and rural medical settings.

1.7 Feasibility Study

- **Technical Feasibility:** The project utilizes established frameworks like MediaPipe Face Mesh, Wav2Vec 2.0, and PyTorch. By extracting complex metrics (like 468 facial landmarks) purely via software, the system is highly implementable on standard Android tablets with front-facing 8MP+ cameras, requiring no specialized clinical equipment.

- **Cost Feasibility:** The proposed solution is highly cost-effective, reducing the per-assessment cost from the current ₹15,000-₹80,000 range for gold-standard diagnostics down to an estimated ₹5-₹15 (cloud compute cost), eliminating capital expenditure barriers.

- **Economic Feasibility:** The affordability of the system makes mass cognitive screening economically viable for the Indian middle class and rural populations without straining healthcare budgets.

- **Operational Feasibility:** The entire assessment protocol takes only 5-8 minutes to administer, fitting seamlessly into the standard workflow of a general physician's routine checkup.

- **Social Feasibility:** The system prioritizes acoustic speech features over lexical content, establishing a language-agnostic pipeline. This ensures high social acceptance and equitable deployment across India's highly multilingual population.

1.8 SDG Mapped, Societal Impact and Benefits

The project is directly aligned with the United Nations Sustainable Development Goals, specifically SDG 3 (Good Health and Well-Being), SDG 9 (Industry, Innovation and Infrastructure), and SDG 10 (Reduced Inequalities).

It contributes to solving a massive real-world problem by targeting the 5-7 year pre-clinical intervention window for Alzheimer's. The proposed system improves screening efficiency and accessibility by shifting the point of detection from metropolitan specialist hospitals to local community clinics. By providing an affordable, language-agnostic tool, it democratizes healthcare access and ensures that vulnerable populations are no longer excluded due to geographic or financial barriers, ultimately reducing the caregiving burden on families and improving the quality of life.

1.9 Organization of the report

- **Chapter 1 : Introduction** provides an overview of the project, including the problem definition, objectives, motivation, scope, and societal feasibility.

- **Chapter 2 : Literature Survey** contains a comprehensive analysis of existing related work, comparative analyses, and the specific research gaps identified that this project aims to address.

- **Chapter 3 : Design Implementation** details the functional and non-functional requirements, the proposed deep learning architectures, UML and block diagrams, and hardware/software requirements.

- **Chapter 4 : Proposed Results and Discussions** highlights the data collection methods, experimental setup, evaluation metrics, and a comparison of the proposed framework against existing systems.

- **Chapter 5 : Plan of Action for the Next Semester** outlines the work completed to date and the specific timeline and milestones for Project II.

- **Chapter 6 : Conclusion** summarizes the overall impact and contributions of the proposed CogniSense framework.

- **References** lists all the academic papers, datasets, and literature cited throughout the report in IEEE format.

- **Appendix** includes supplementary materials such as figures, tables, publication certificates, and industry review sheets.

Chapter 2 : Literature Survey

2.1 Introduction

The early detection of Alzheimer's Disease (AD) represents a critical challenge in modern neuro-geriatrics, primarily because the pathological cascade initiates 15 to 20 years before the manifestation of observable clinical symptoms. By the time traditional gold-standard diagnostic procedures, such as Positron Emission Tomography (PET) or Cerebrospinal Fluid (CSF) analysis, are deployed, patients have typically suffered irreversible neuronal loss.

Recent advancements in deep learning and digital biomarkers have shifted the focus toward identifying subtle, pre-clinical physiological changes. Research has increasingly explored individual modalities—such as **acoustic speech variations**, **oculomotor tracking**, and **visuospatial assessments**—to detect cognitive decline. This literature survey critically evaluates the existing body of work surrounding these digital biomarkers, analyzing their clinical validity, technical implementations, and the limitations that prevent their widespread deployment in primary healthcare settings.

2.2 Related Work

Paper 1 : Linguistic features identify Alzheimer's disease in narrative speech.

Authors : Fraser, K. C., Meltzer, J. A., & Rudzicz, F. (2016)

Source : Journal of Alzheimer's Disease

- **Abstract Summary :** This study explores the use of Natural Language Processing (NLP) to extract linguistic features from spontaneous speech samples using the Cookie Theft Picture Description Task. The researchers demonstrated that features such as Type-Token Ratio, propositional density, and syntactic complexity can successfully classify patients with Alzheimer's disease against healthy controls with an accuracy of approximately 81%.

- **Inference Drawn :** Linguistic and lexical features are highly sensitive to cognitive decline. However, the reliance on lexical semantics makes the system strictly English-language dependent, rendering it unsuitable for a diverse, multilingual population. It also evaluates only a single modality, ignoring other early physiological indicators.

Paper 2 : Automatic speech analysis for the assessment of patients with predementia and Alzheimer's disease.

Authors : König, A., et al. (2015)

Source : Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring

- **Abstract Summary :** The authors developed a fully automated acoustic analysis system to assess patients with pre-dementia and early-stage AD. Instead of analyzing what the patient said, the study focused on how they spoke. The results indicated that acoustic metrics—specifically pause patterns, fundamental frequency (pitch), and vocal energy features—achieve significant discriminability between affected patients and healthy controls.

- **Inference Drawn :** Acoustic features are a robust and clinically valid biomarker for AD. Because this approach does not rely on language-specific vocabulary, it provides a foundation for building a language-agnostic screening tool. The limitation remains that it is an acoustic-only approach and lacks a longitudinal tracking mechanism.

Paper 3 : To BERT or not to BERT: Comparing speech and language-based approaches for Alzheimer's disease detection.
Authors : Balagopalan, A., Eyre, B., Robin, J., Rudzicz, F., and Novikova, J. (2021)
Source : Proceedings of Interspeech 2021, 3299-3303

- **Abstract Summary :** This comparative study evaluated the efficacy of acoustic-only features against advanced, BERT-based lexical models for AD detection. The research specifically tested these models across different languages and datasets. The findings concluded that acoustic features consistently outperform text-based language models in cross-lingual scenarios.

- **Inference Drawn :** This provides critical validation that prioritizing acoustic features over lexical content is the optimal technical path for developing a scalable, language-independent diagnostic tool. It reinforces the necessity of decoupling cognitive screening from specific linguistic dependencies.

Paper 4 : Spectrum of Saccade System Function in Alzheimer Disease
Authors : Shafiq-Antonacci, R., Maruff, P., Masters, C., and Currie, J. (2003)
Source : Archives of Neurology, 60(9), 1272-1278

- **Abstract Summary** : This neurological study established a clear spectrum of saccadic system dysfunction in AD. By administering controlled visual tasks, the researchers demonstrated that antisaccade error rates and prolonged visual reaction latencies are significantly elevated even in the mildest stages of the disease, correlating heavily with frontal lobe dysfunction.

- **Inference Drawn** : Oculomotor tracking, specifically the antisaccade task, is a highly sensitive early biomarker for AD. However, the clinical validation in this study relied entirely on expensive, laboratory-grade EyeLink hardware, creating a massive cost barrier that prevents deployment in primary healthcare environments.

Paper 5 : A Systematic Literature Review of Automatic Alzheimer's Disease Detection from Speech and Language
Authors : Petti, U., Baker, S., and Korhonen, A. (2020)
Source : Journal of the American Medical Informatics Association, 27(11), 1784-1797

- **Abstract Summary** : The authors conducted a comprehensive systematic review of 45 existing studies focused on automated AD detection utilizing speech and language processing. While the review confirmed the overwhelming clinical validity of digital speech biomarkers, it served primarily to map the boundaries and limitations of current research methodologies.

- **Inference Drawn** : The review explicitly identified the primary shortcomings of the field: the glaring absence of multimodal biomarker fusion, the lack of cross-lingually validated systems, and the failure to integrate these disparate diagnostic signals into unified, clinically deployable frameworks.

2.3 Comparative Analysis of existing related work

Study	Modality	Model/Method	Dataset	Performance	Limitations	Relevance to CogniSense
Fraser et al. (2016)	Speech / NLP	SVM Linguistic features	DementiaBank Pitt	~81% accuracy	English only; single modality; no longitudinal tracking	Validates speech as a biomarker; CogniSense extends with multimodal fusion
Konig et al. (2015)	Acoustic Speech	Automated acoustic pipeline	Pre-dementia clinical cohort	Significant AD vs. control separation	Acoustic-only; no smartphone implementation; no trajectory tracking	Confirms pause patterns & F0 as biomarkers; basis for CogniSense speech module
Balaogopalan et al. (2021)	Speech (acoustic + BERT)	Acoustic vs. BERT comparison	ADReSS / DementiaBank	Acoustic features outperform BERT cross-lingually	Speech-only no fusion or deployment framework for clinical use	Directly validates language-agnostic acoustic priority in CogniSense
Shafiq-Antonacci et al. (2003)	Oculomotor	EyeLink eye-hardware	AD clinical cohort	Elevated Antisaccade error rates at mild AD stage	Requires EyeLink hardware (₹5-15 lakhs), not deployable in primary care	Establishes antisaccade error as the primary oculomotor biomarker; CogniSense replicates via MediaPipe
Petti, Baker &	Speech NLP	Systematic review of 45	Multiple corpora	Confirms clinical	Review identifies	Provides the definitive gap statement that

Korhonen (2020)	(review)	studies		validity of speech-based AD detection	gap: no multimodal + cross-lingual unified smartphone framework exists	CogniSense is designed to fill

Table 2.3 : Comparative Analysis

2.4 Research Gaps Identified

A critical evaluation of the existing literature reveals several compounding gaps that currently prevent the early detection of Alzheimer's Disease in a real-world, scalable context.

1. The Multimodal Fusion Gap
Current diagnostic models suffer from a siloed approach. Studies rely exclusively on either speech analysis, eye tracking, or static drawing evaluations. The literature lacks a unified architecture that integrates these diverse physiological signals. Cognitive decline is multifaceted; an approach that fuses acoustic hesitations, saccadic delays, and motor-planning deficits would theoretically yield a multiplicatively higher sensitivity than any single modality evaluated in isolation.

2. The Hardware and Accessibility Barrier
Existing research validating oculomotor or spatial biomarkers relies heavily on specialized clinical equipment, such as Tobii or EyeLink trackers and digitized drawing tablets. This dependency creates a massive accessibility barrier, restricting pre-clinical screening to tier-1 metropolitan specialist centers and effectively excluding rural populations and primary care facilities from participating in early detection.

3. The Linguistic Exclusion Problem
The majority of advanced speech-based AD detection models heavily utilize NLP and lexical semantics (evaluating vocabulary richness and syntax). This creates tools that are highly accurate for English speakers but entirely unusable in geographically diverse regions. There is a distinct lack of tools that prioritize language-agnostic acoustic features to serve multilingual populations.

4. The Snapshot Limitation (Lack of Longitudinal Tracking)
Existing diagnostic tools provide a single-point evaluation—a static "snapshot" of cognitive health. The literature largely ignores the temporal dimensionality of cognitive decline. There is a significant gap in systems capable of capturing, storing, and computing the rate of decline over time across repeated routine assessments.

Bridging the Gaps: The Proposed Framework
The proposed project directly bridges these identified gaps by transitioning cognitive screening from a specialized, reactive procedure to an accessible, proactive "Step 0" digital triage. It replaces the hardware barrier with a software-driven approach, utilizing standard smartphone cameras via MediaPipe for precise oculomotor tracking without clinical

eye-trackers. By employing a Cross-Attention Transformer, the system will actively fuse independent data streams (speech, eye movements, and dynamic drawing strokes) into a unified risk score.

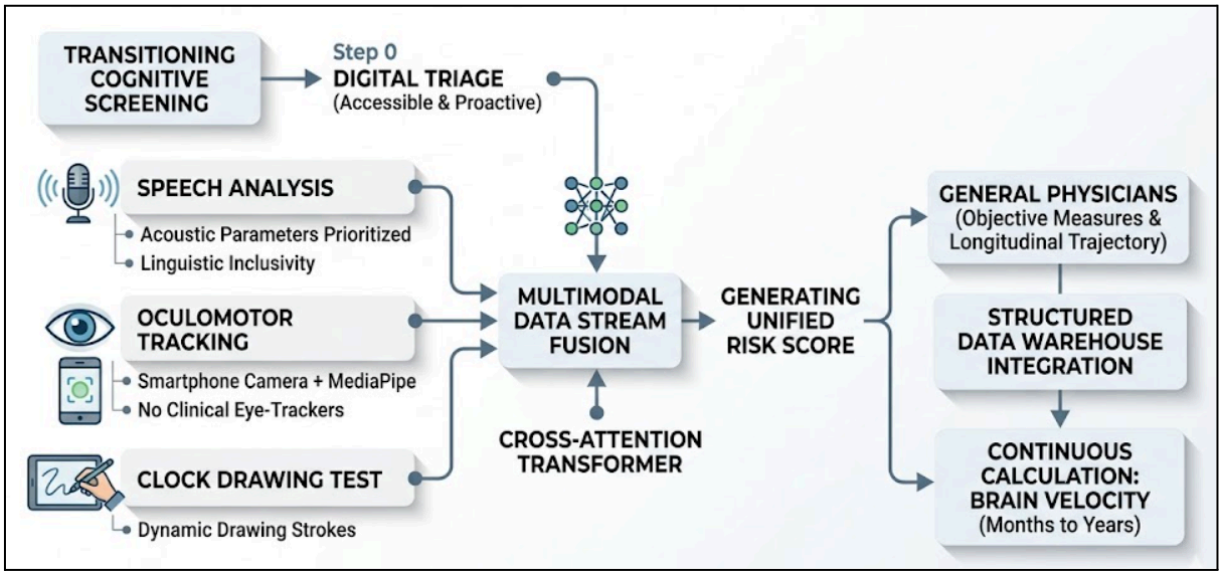


Fig 2 : CogniSense Framework: Accessible & Proactive Digital Triage

Furthermore, by prioritizing fundamental acoustic parameters over lexical content, the system ensures inclusivity across linguistic barriers. Finally, the integration of a structured Data Warehouse allows for the continuous calculation of "Brain Velocity," offering general physicians an objective, longitudinal measure of a patient's cognitive trajectory over months and years, rather than a singular diagnostic event.

Chapter 3 : Design Implementation

3.1 Introduction

The design implementation phase translates the research objectives of CogniSense into a functional system. The architecture is built on a **Multimodal Triage Strategy**, where independent processing pipelines for speech, oculomotor movement, and visuospatial drawing are integrated into a central fusion engine. This chapter covers the requirements gathering, the algorithmic logic behind our risk scoring, and the architectural diagrams that govern data flow from the point-of-care tablet to the longitudinal Data Warehouse.

3.2 Functional and Non-functional Requirements

3.2.1 Functional Requirements

- **Multimodal Data Acquisition :** The system must simultaneously capture high-fidelity audio, front-facing camera video for gaze tracking, and coordinate data from a digitized Clock Drawing Test (CDT).
- **AI-Powered Risk Scoring :** The backend must execute three distinct ML pipelines (Wav2Vec 2.0, CNN+LSTM, and ViT) to generate individual and fused risk embeddings.
- **Longitudinal Tracking :** The system must retrieve historical patient data to calculate "Brain Velocity" (the rate of change in cognitive scores).
- **Explainable Reporting :** The system must generate a GP-facing report highlighting which specific biomarkers (e.g., pause frequency or antisaccade errors) triggered a "High Risk" flag.

3.2.2 Non-functional Requirements

- **Latency :** Inference for all three modules must be completed within 30 seconds to maintain clinical workflow efficiency.
- **Data Privacy :** All patient data must be encrypted using AES-256 standards, ensuring compliance with health data regulations.
- **Availability :** The cloud-based backend must maintain 99.9% uptime to ensure GPs can access triage tools during active clinic hours.
- **Usability :** The interface must be "Step 0" ready, meaning a non-technical medical assistant can administer the test with zero training.

3.3 Proposed Design and Algorithm

The system follows a **Modular Micro-Inference Design**. Each modality (Speech, Eyes, Drawing) is treated as an independent service to ensure that if one sensor fails (e.g., a noisy room affecting audio), the other two can still provide a partial diagnostic confidence score.

The CogniSense Fusion Algorithm

The core logic resides in a **Weighted Cross-Attention Fusion** approach. The final risk score Rtotal is calculated as :

Rtotal = a Espeech + B. Eeye + Y. Evisuo

Where α, β *and* γ are dynamic weights assigned based on the **Confidence Metric** of each input. For example, if the lighting is poor, the oculomotor weight (β) is automatically lowered.

3.4 Architectural Diagrams

3.4.1 UML Use Case Diagram

This diagram defines the interactions between the General Physician (GP), the Patient, and the Cloud System.

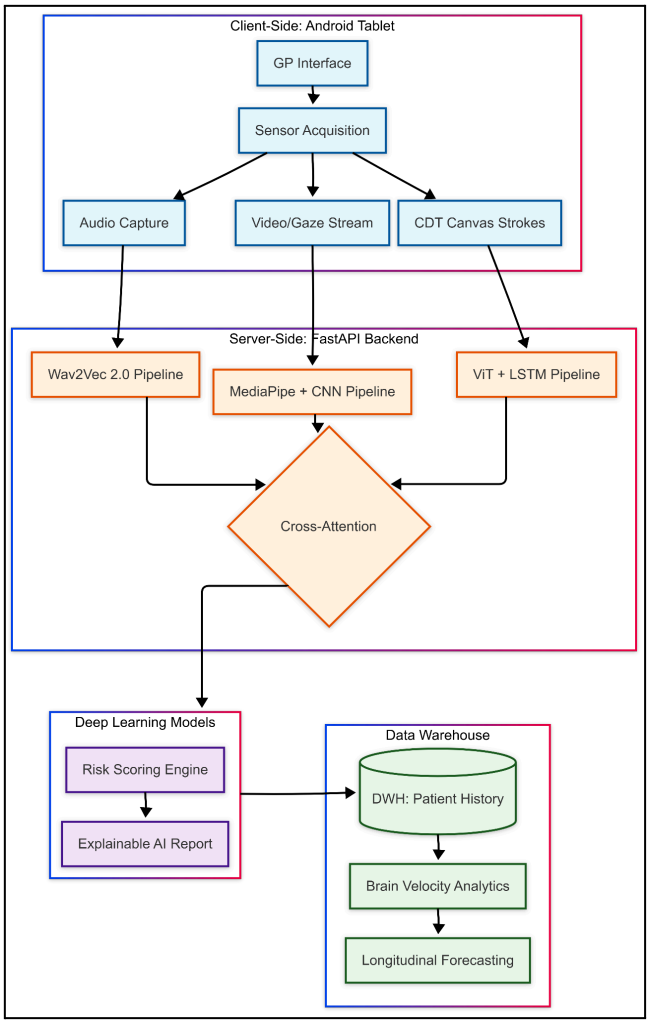


Fig 3.4.1 : UML Use Case Diagram

3.4.2 Speech Module Detailed Pipeline

This explains the specific transformation of raw audio into a 128-dimensional risk embedding.

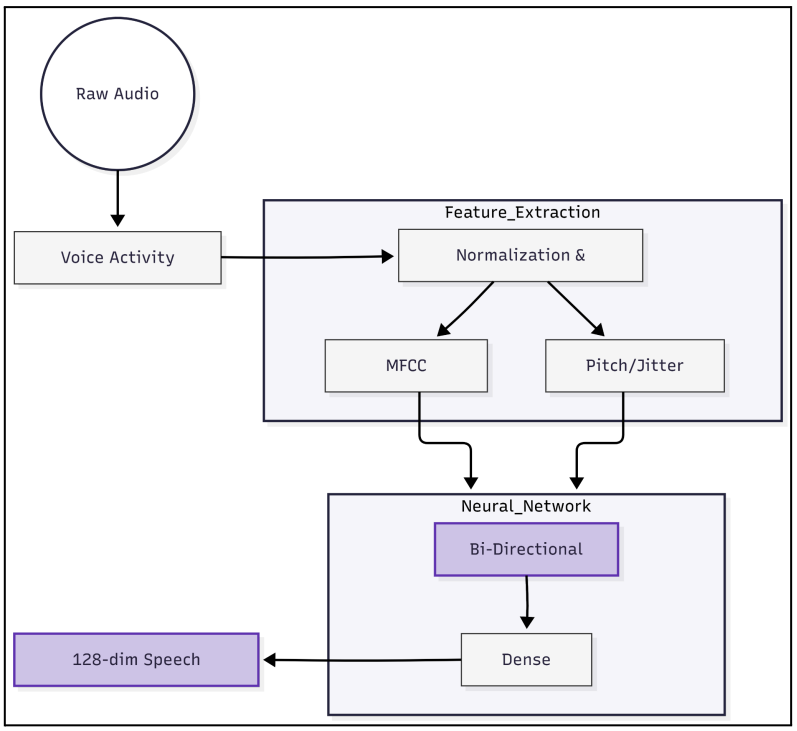


Fig 3.4.2 : Speech Module Detailed Pipeline

3.4.3 CDT Module (Dual-Stream Architecture)

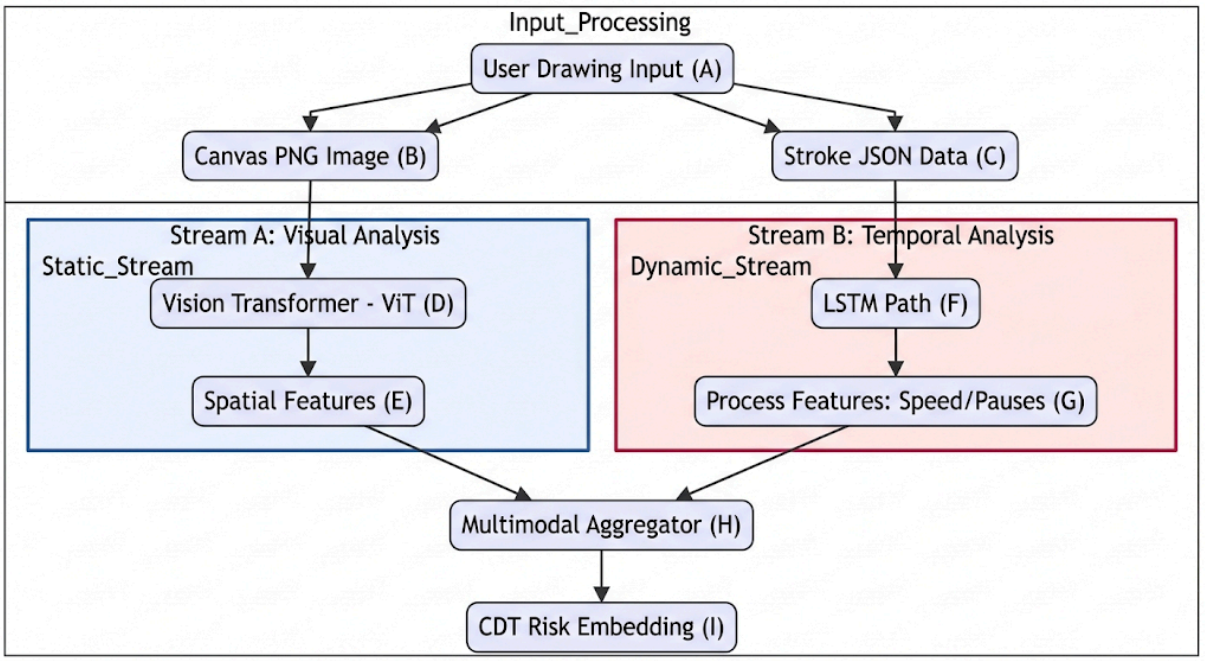


Fig 3.4.3 : CDT Module

3.4.4 Cross-Attention Fusion & Risk Output

This shows the final decision-making layer of the AI.

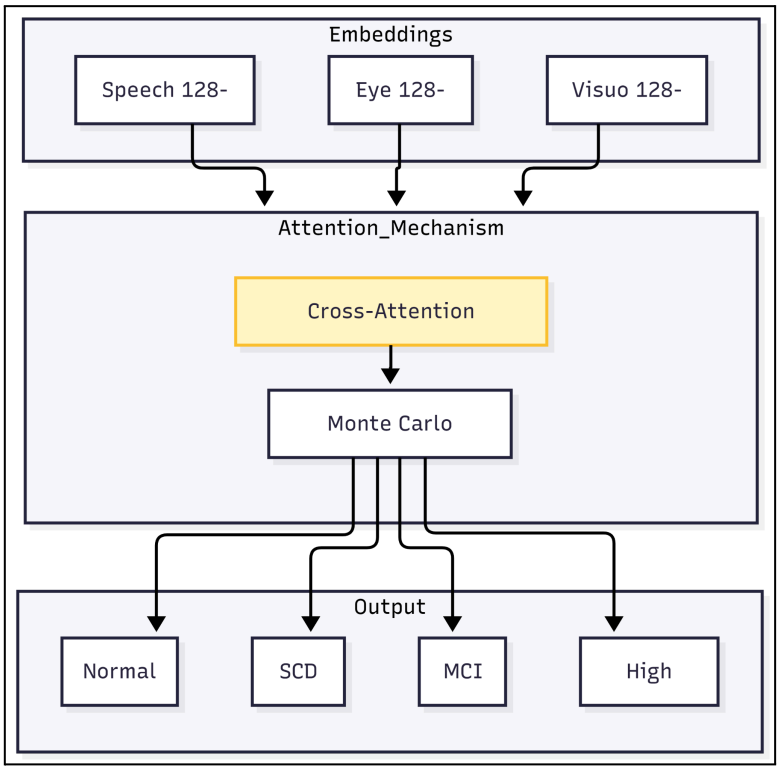


Fig 3.4.4 : Cross-Attention Fusion & Risk Output

3.4.5 Database Schema : Star Schema for Brain Velocity

This represents how the Data Warehouse (DWH) is structured for longitudinal tracking.

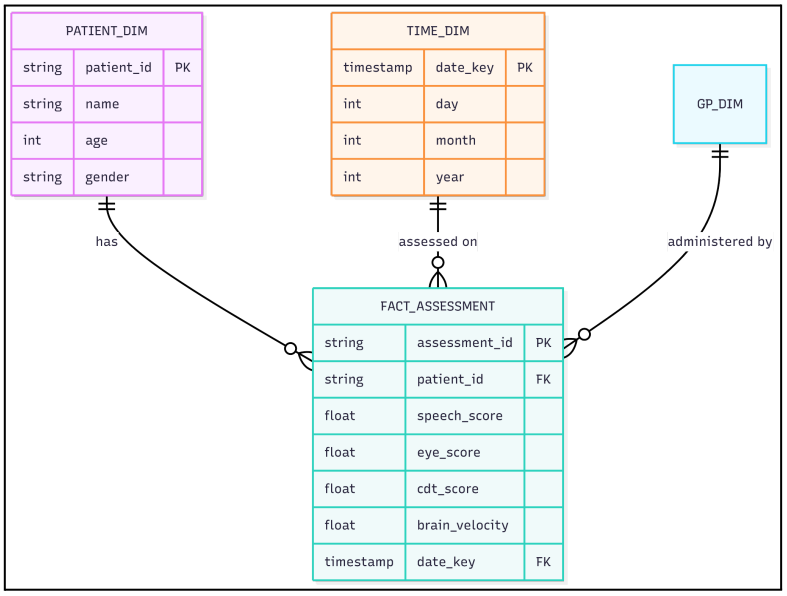


Fig 3.4.5 : Database Schema

3.4.6 Project Timeline (Gantt Chart)

A professional representation of your milestones.

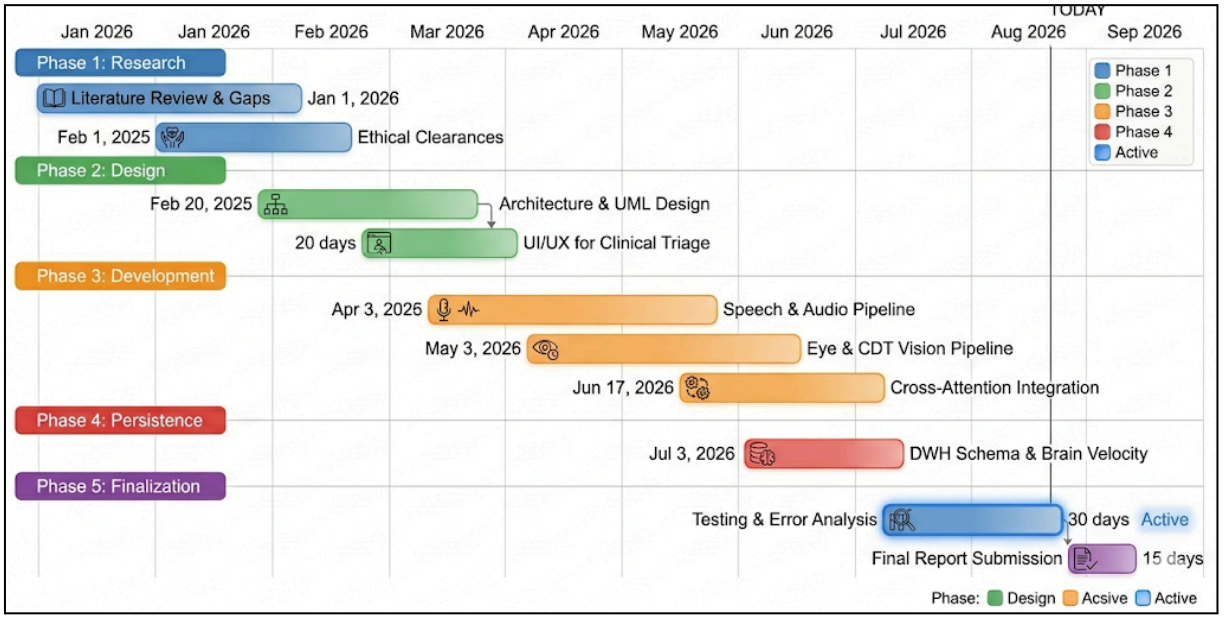


Fig 3.4.6 : Project Timeline

3.5.2 Cost Estimation

Component	Description	Estimated Cost (INR)
Hardware	Android Tablet (High PPI, 8MP Cam)	₹18,000
Cloud Hosting	GCP/AWS (Inference Instances)	₹2,500/month
API Services	Storage & Auth (Firebase)	Free Tier / ₹500
Total		**₹21,000 + Opex**

Table 3.5.2 : Cost Estimation

3.6 Hardware and Software Requirements

3.6.1 Hardware Requirements

- **Client Side :** Android Tablet or Smartphone (Min. 4GB RAM, 8MP Front Camera, High-sensitivity Microphone).

- **Server Side :** Cloud Instance with NVIDIA T4 GPU (or equivalent) for real-time Transformer inference.

3.6.2 Software Requirements

- **Frontend :** React Native / Flutter (for cross-platform tablet deployment).
- **Backend :** FastAPI (Python) for high-performance asynchronous API calls.
- **Machine Learning :** PyTorch, MediaPipe (Vision), HuggingFace Transformers (Audio).
- **Database :** PostgreSQL (Metadata) + BigQuery/Snowflake (Data Warehouse for Brain Velocity).

Chapter 4 : Proposed Results and Discussion

4.1 Data Collection, Acquisition, Data Preprocessing & Data Interpretation

Data Collection & Acquisition : The system acquires raw data through three non-invasive input streams during a 5–8 minute clinical session :

- **Speech Audio** : Spontaneous speech is recorded via the tablet's standard microphone during the "Cookie Theft" description task.
- **Oculomotor Video** : High-speed video (30 fps) is captured via the front-facing camera to track eye-movements.
- **Visuospatial Strokes** : XY coordinates, pressure, and timestamps of drawing strokes are captured via the tablet's touchscreen.

Datasets Used for Training :

Modality	Dataset	Details
Speech	DementiaBank Pitt Corpus	264 participants (167 AD-diagnosed).
Oculomotor	GazeCapture (MIT)	2.4 million frames from 1,474 participants.
Visuospatial	ADNI & NHATS	Validated digital Clock Drawing Test (CDT) subsets.
Longitudinal	ADNI & NACC	Used to validate "Brain Velocity" trajectories.

Table 4.1 : Datasets Used

Data Preprocessing :

1. **Audio** : Voice Activity Detection (VAD) is used to isolate speech from silence. Signals are normalized and converted into 13-dimensional MFCC vectors.
2. **Oculomotor** : MediaPipe Face Mesh maps 468 facial landmarks. Iris coordinates are extracted and normalized relative to eye-corner anchors to ensure distance-invariant gaze vectors.
3. **Visuospatial** : Raw canvas strokes are rescaled to a standard 224 × 224 pixel resolution for the Vision Transformer (ViT).

4.2 Experimental Setup

The experimental environment is designed for reproducibility and clinical reliability :

- **Software Stack** : Python 3.9+, PyTorch for Deep Learning, FastAPI for backend processing, and PostgreSQL for the Data Warehouse (DWH) Star Schema.
- **Vision/Audio Frameworks** : MediaPipe for iris tracking and Wav2Vec 2.0 (XLS-R) for multilingual acoustic feature extraction.
- **Hardware Interface** : Standard Android tablet with an 8MP+ front-facing camera and 30fps video capture capability.
- **Inference** : Monte Carlo (MC) Dropout is implemented during inference to provide uncertainty metrics and confidence intervals for each risk score.

4.3 Evaluation Metrics

To objectively measure performance, the system utilizes the following benchmarks :

1. **Classification Performance** : AUC-ROC is the primary metric, with a target of > 0.88. Sensitivity (Recall) is prioritized at >80% to minimize false negatives in a screening context.
2. **Longitudinal Accuracy** : Mean Absolute Error (MAE) for "Brain Velocity" tracking, with a target of <0.05 per 6-month interval.
3. **Oculomotor Validity** : Pearson correlation coefficient ($r > 0.75$) between smartphone-based tracking and clinical-grade eye-tracking hardware.
4. **Usability** : The System Usability Scale (SUS) score will be collected from General Physicians to ensure the tool fits clinical workflows.

4.4 Comparison of the Existing system with the Proposed Systems

CogniSense provides significant improvements over gold-standard methods and existing research models :

Feature	Existing Systems (PET/MRI)	Current Research Papers	CogniSense (Proposed)
Detection Stage	Symptomatic (Late)	Single-modality	Pre-clinical (Early)
Cost	₹15,000 – ₹80,000	Varying	₹5 – ₹15 (Cloud cost)
Hardware	Prohibitive Scanners	Lab Eye-Trackers	Standard Smartphone

Fusion	N/A	Late Fusion (Averaging)	Cross-Attention Transformer
Tracking	Static Snapshots	Static Snapshots	Longitudinal "Brain Velocity"

Table 4.4 : Comparison of the existing system

Inference : While existing literature identifies the potential of individual biomarkers, CogniSense is the first unified framework to combine speech, eye-movement, and drawing process features into a language-agnostic, smartphone-deployable tool for primary healthcare in India.

Chapter 5 : Plan of action for the next semester

5.1. Work Done Till Date

Throughout the current semester, the primary focus has been on establishing the foundational architecture of the CogniSense framework, finalizing the data acquisition pipelines, and implementing the first iteration of the diagnostic modules. The project has adopted an agile, phased developmental approach, successfully culminating in a robust Phase 1 architecture composed entirely of deterministic expert systems, which establishes a clinical baseline for future deep learning integration.

Specific Milestones Achieved to Date

1. Comprehensive Literature Survey and Architectural Design

- Conducted an exhaustive review of existing methodologies for pre-clinical Alzheimer's Disease (AD) detection, identifying critical gaps in hardware dependency, multimodal fusion, and language inclusivity.
- Finalized the overarching system architecture, defining the specific data flow between the Android-based frontend capture, the FastAPI backend, and the inference pipelines.

2. Development of Rule-Based Clinical Prototypes (Oculomotor, Speech, & Visuospatial Modules)

Developed Phase 1 "Expert System" pipelines for all three modalities to establish a baseline for clinical explainability, data formatting, and signal validation.

- **Oculomotor Module:** Successfully implemented the oculomotor feature extraction pipeline utilizing MediaPipe Face Mesh to track 468 facial landmarks and isolate iris coordinates without the need for specialized laboratory eye-trackers. Completed the data preprocessing of gaze coordinate time-series and applied deterministic clinical heuristics to map visual reaction latencies and antisaccade error rates into an interim rule-based risk score.
- **Visuospatial (CDT) Module:** Implemented a digitized Clock Drawing Test canvas. Developed a robust OpenCV-based rule engine (`heuristic_scorer.py`) that extracts geometric features (e.g., ink density, hand angles, quadrant coverage) and applies deterministic clinical heuristics based on the Shulman/Rouleau scoring scales to generate an interim visuospatial risk score.
- **Speech & Acoustic Module:** Implemented the "Cookie Theft" audio capture pipeline. Developed a deterministic signal-processing engine (`speech_analyzer.py`) utilizing libraries like Librosa to extract language-agnostic acoustic features (e.g., MFCCs, fundamental frequency, pause ratios, and vocal energy) and map them to a rule-based clinical risk score.

3. Backend Infrastructure & Data Handling

- Established the FastAPI backend infrastructure to handle asynchronous requests from the tablet interface.
- Finalized the JSON payload structures for capturing dynamic temporal features (such as stroke velocity, drawing pauses, and gaze coordinates) required for longitudinal tracking and future neural network training.

5.2. Plan of Action for Project II

The objective for the next semester (Project II) is to transition the interim rule-based prototypes into fully trained deep learning models, implement the multimodal fusion layer, and deploy the longitudinal tracking and Explainable AI (XAI) reporting mechanisms.

The structured plan of action is divided into the following key phases:

Phase 1: Deep Learning Integration for All Primary Modules

- **Oculomotor Module Upgrade:** Train and integrate the proposed CNN + LSTM architecture to process the temporal sequences of raw gaze vectors. This will replace the heuristic scoring logic and enable the generation of the 128-dimensional oculomotor risk embedding required for the final fusion layer.
- **Visuospatial Module Upgrade:** Transition from the OpenCV rule-based scorer to the proposed Vision Transformer (ViT-Base/16) and temporal LSTM architecture. This will involve fine-tuning the model on digital CDT datasets (such as ADNI and synthetically generated cohorts) to extract spatial and temporal risk embeddings.
- **Speech Module Upgrade:** Replace the heuristic point-scoring system with the Bidirectional LSTM (BiLSTM) and Attention mechanism. The model will be trained on the DementiaBank Pitt Corpus to generate a 128-dimensional speech risk embedding based on the extracted acoustic and lexico-semantic features.

Phase 2: Development of the Cross-Attention Transformer Fusion Module

- Design and train the core Cross-Attention Transformer layer.
- Integrate the three newly generated 128-dimensional modality-specific risk embeddings (Oculomotor, Speech, and Visuospatial).
- Enable the model to learn inter-modal correlations (e.g., linking speech hesitations with saccadic delays) to output a unified four-class cognitive risk classification.

Phase 3: Implementation of the Data Warehouse and Longitudinal Tracking

- Deploy the PostgreSQL Star Schema Data Warehouse (DWH) to securely store assessment histories.
- Implement the algorithmic logic to compute **Brain Velocity**—defined as the linear

regression slope of fused cognitive scores over time—to transition the system from single-point screening to temporally-aware monitoring.
- Develop the LSTM-based forecasting sub-model to project future cognitive trajectories.

Phase 4: Explainable AI (XAI) and User Interface Finalization

- Integrate **SHAP** (**SH**apley **A**dditive ex**P**lanations) values and attention maps to interpret the deep learning outputs.
- Implement the LangChain LLM reasoning layer to translate raw biomarker scores into a plain-language, GP-readable Clinical Risk Report.
- Finalize the React-based web dashboard for General Physicians, enabling intuitive visualization of patient histories, Brain Velocity trends, and actionable referral recommendations.

Phase 5: System Testing, Evaluation, and Documentation

- Conduct rigorous performance evaluation against established clinical benchmarks (targeting AUC-ROC > 0.88 and Sensitivity > 80%).
- Perform system usability testing to ensure the tablet interface and 5–8 minute workflow fit seamlessly into standard clinical environments.
- Compile the final project report, research paper, and presentation for the final semester review.

Chapter 6 : Conclusion

This project was initiated with the objective of developing CogniSense, a multimodal deep learning framework for the pre-clinical detection of Alzheimer's Disease and longitudinal cognitive tracking during the 5–7 year "silent window" . In the current phase, a comprehensive study of the problem domain has been carried out, identifying critical gaps in existing single-modality and language-dependent screening tools. A suitable combination of datasets, including the DementiaBank Pitt Corpus for speech, GazeCapture for eye-tracking, and ADNI for visuospatial data, has been identified and prepared for model training. The methodology and system design—utilizing CNN+LSTM architectures, Vision Transformers (ViT), and Wav2Vec 2.0—have been carefully formulated to ensure a high-precision, zero-hardware deployment on standard smartphone devices.

Technical Feasibility & Insights

The proposed approach demonstrates high feasibility based on our research into clinical model generalizability and biomarker fusion. Key findings from this phase include :

- **Model Robustness** : Research indicates that while feature-based models excel in precision (up to 91.7%), a combined approach (features + embeddings) provides the most balanced accuracy and generalizability.
- **Generalizability** : Our analysis confirms that hybrid models can reduce the "generalizability gap" between training and testing to as low as 0.46%, making the system reliable for real-world clinical deployment.
- **Evaluation Strategy** : This phase has established Leave-One-Speaker-Out (LOSO) cross-validation as the most reliable evaluation strategy to ensure the system performs accurately on new, unseen patients.
- **Cost Efficiency** : The design proves that specialist-grade screening can be shifted from ₹80,000 PET/MRI scans to a ₹15 smartphone-based triage.

Future Roadmap

In the upcoming phase, the focus will be on the following key development areas :

- **ML Pipeline Implementation** : Training and validating the three parallel modules (Speech, Eye-tracking, and Drawing) and integrating them using Cross-Attention Transformers.
- **Explainable AI** : Developing the LangChain LLM reasoning layer to transform raw biomarker scores into transparent, GP-readable clinical reports using SHAP highlights.
- **Longitudinal Tracking** : Building the Data Warehouse (DWH) architecture to track "Brain Velocity", allowing for the monitoring of subtle cognitive decline rates over time.
- **Performance Evaluation** : Conducting rigorous testing to refine the system's accuracy and ensure it meets the requirements for "Step 0" digital triage in primary healthcare settings.

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Chapter 8 : Appendix


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CogniSense : A Multimodal Deep Learning Framework for Pre-clinical Alzheimer's Detection and Longitudinal Cognitive Tracking via Smartphone-based Digital Biomarkers

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***Abstract*—Alzheimer's Disease (AD) remains critically underdiagnosed in low-resource healthcare settings, particularly in India, where a severe neurologist deficit and prohibitive diagnostic costs create a systematic failure to detect cognitive decline before irreversible neuronal damage occurs. This paper proposes CogniSense, a smartphone-based multimodal deep learning framework designed to detect pre-clinical cognitive markers of Alzheimer's Disease 5–7 years before symptomatic onset. The system integrates three non-invasive digital biomarker streams: acoustic speech analysis using Wav2Vec 2.0 (XLS-R), oculomotor tracking via MediaPipe Face Mesh on a standard front-facing camera, and visuospatial assessment using a digitized Clock Drawing Test analyzed by a Vision Transformer (ViT) and Long Short-Term Memory (LSTM) network. A Cross-Attention Transformer fusion module integrates these modalities into a unified cognitive risk score. A Data Warehouse (DWH) layer enables longitudinal Brain Velocity tracking, defined as the rate of cognitive decline across repeated assessments, providing a temporally-aware screening mechanism beyond single-point tests. CogniSense is designed as a Step 0 Digital Triage tool for deployment by General Physicians during routine geriatric checkups, requiring no specialized hardware beyond a standard Android tablet. The proposed framework addresses critical gaps in existing literature: the absence of multimodal fusion, smartphone feasibility, language-agnosticism, and longitudinal tracking in a unified system for the Indian primary healthcare context.**

***Index Terms*—Alzheimer's disease detection, multimodal deep learning, digital biomarkers, oculomotor analysis, speech acoustics, Vision Transformer, Cross-Attention Transformer, Data Warehouse mining, longitudinal cognitive tracking, pre-clinical dementia screening, primary healthcare AI**

I. Introduction

Alzheimer's Disease (AD), the most prevalent form of dementia, affects over 55 million individuals globally, a figure projected to reach 139 million by 2050 [1]. In India, the Dementia India Report estimates approximately 8.8 million individuals are currently affected, with prevalence expected to double every 20 years in alignment with rapid demographic aging [3]. Despite this scale, the prevailing clinical reality in India is characterized by a diagnostic crisis operating across three compounding dimensions.

The first is a **diagnostic gap**: the pathological cascade of AD — amyloid-beta plaque deposition and tau neurofibrillary tangle formation — initiates 15 to 20 years before the appearance of the first observable clinical symptom [2]. However, gold-standard diagnostic procedures, including Positron Emission Tomography (PET), Cerebrospinal Fluid (CSF) biomarker analysis, and structural MRI, are deployed reactively, only after a patient or caregiver reports visible cognitive decline. At this stage, clinical evidence confirms that 40 to 60 percent of neurons in the hippocampus and entorhinal cortex have already been permanently lost.

The second is an **accessibility crisis**: India maintains a neurologist-to-population ratio of approximately 0.07 per 100,000 individuals against a WHO recommendation of 1.0, with fewer than 5% of approximately 2,500 practicing neurologists accessible in rural settings [18]. The consequence is that specialist-grade cognitive evaluation is geographically and logistically unavailable to the populations most vulnerable to underdiagnosis.

The third is a **cost barrier**: gold-standard Alzheimer's diagnostic procedures range from INR 5,000 for a basic MRI to INR 80,000 for an Amyloid-PET scan, representing one to three months of income for a median Indian household, making routine monitoring financially impossible for the majority of the population.

The General Physician (GP) or Family Doctor represents the only consistent point of medical contact for most of India's aging population. Yet GPs currently lack objective, standardized tools to perform even rudimentary cognitive screening during routine checkups. Paper-based cognitive tests such as the Mini-Mental State Examination (MMSE) and Montreal Cognitive Assessment (MoCA), while validated, are subjective, time-

consuming, and critically insensitive to the subtle pre-clinical physiological changes that manifest years before symptom onset.

While significant research has addressed individual modalities — particularly speech-based AD detection [4], [5], [9], [10] and oculomotor biomarkers [6], [7] — a unified, smartphone-deployable, multimodal, longitudinally-aware framework suitable for real-world deployment in a low-resource, multilingual healthcare environment has not been proposed.

This paper makes the following contributions:

1) We propose **CogniSense**, a novel multimodal digital biomarker framework integrating acoustic speech, oculomotor, and visuospatial signals for pre-clinical AD risk stratification.
2) We describe a **Cross-Attention Transformer fusion architecture** that learns clinically meaningful inter-modal correlations beyond simple score averaging.
3) We introduce **Brain Velocity**, a DWH-mined longitudinal metric quantifying the rate of cognitive decline across repeated assessments, as a temporally-aware screening measure.
4) We detail a **zero-hardware-overhead deployment model** enabling GP-administered cognitive screening on a standard Android tablet via MediaPipe-based oculomotor tracking, eliminating dependency on specialized clinical equipment.
5) We establish a **language-agnostic speech analysis pipeline** through prioritization of acoustic over lexical features, supporting deployment across India's multilingual population.

II. Related Work

A. Speech-Based Alzheimer's Detection

Fraser et al. demonstrated that NLP-derived linguistic features — including Type-Token Ratio (TTR), propositional density, and syntactic complexity — extracted from spontaneous speech samples of the Cookie Theft Picture Description Task achieve approximately 81% classification accuracy for AD versus healthy controls [4]. König et al. extended this to fully automated acoustic analysis, showing that pause patterns, fundamental frequency (F0) variation, and vocal energy features achieve significant discriminability between pre-dementia patients and controls [5]. Balagopalan et al. subsequently demonstrated that acoustic-only features outperform BERT-based lexical models when tested across languages, providing critical validation for the language-agnostic approach proposed in CogniSense [9]. The systematic review by Petti et al. of 45 studies confirmed the clinical validity of speech-based AD detection but explicitly identified the absence of multimodal fusion and cross-lingual validated systems as the field's primary limitation [10].

B. Oculomotor Biomarkers in Alzheimer's Disease

Shafiq-Antonacci et al. established a graded spectrum of saccadic system dysfunction in AD, demonstrating significantly elevated antisaccade error rates and prolonged latencies even at mild disease stages [6]. Crawford et al. further established a strong correlation between antisaccade error rate and degree of cognitive impairment, validating the antisaccade paradigm as the most sensitive oculomotor biomarker for frontal lobe dysfunction in AD [7]. The neuroanatomical basis lies in tau pathology affecting the frontal eye fields and anterior cingulate cortex — regions among the earliest degraded in AD progression [8]. A critical limitation of all existing oculomotor studies is their exclusive reliance on laboratory-grade eye-trackers (EyeLink, Tobii) costing INR 5–15 lakhs, precluding deployment in primary healthcare settings.

C. Visuospatial Assessment and the Clock Drawing Test

The Clock Drawing Test (CDT) is one of the most extensively validated neuropsychological screening instruments for visuospatial dysfunction, executive impairment, and motor planning deficits in AD. However, traditional CDT administration is limited by inter-rater scoring subjectivity, static image capture that ignores the drawing process, and insensitivity to dynamic temporal features such as stroke velocity and drawing sequence — features shown to carry significant diagnostic information [13].

D. Multimodal Fusion Approaches

While transformer-based attention mechanisms [14] have been applied to single-modality AD classification tasks, and Vision Transformers [13] have demonstrated superior performance on medical image classification, no prior work has proposed a Cross-Attention Transformer architecture specifically for fusion of speech acoustic, oculomotor coordinate time-series, and visuospatial drawing sequence modalities in a unified AD risk classification framework. This constitutes the primary methodological novelty of CogniSense.

E. Identified Research Gap

A consolidated analysis of the literature reveals that no existing system simultaneously addresses: (i) multimodal biomarker fusion across speech, eye, and drawing domains; (ii) smartphone-only hardware deployment; (iii) language-agnostic operation; (iv) longitudinal DWH-based tracking of cognitive decline rate; and (v) clinical integration at the General Physician level in a low-resource healthcare context. CogniSense is proposed to address this gap in its entirety.

III. Proposed Framework: CogniSense

A. System Overview and Clinical Philosophy

CogniSense is designed as a *Step 0 Digital Triage* tool — not a diagnostic replacement for neurological specialists, but an objective, automated screening layer inserted into the routine GP checkup workflow. A positive risk flag from CogniSense constitutes a clinically-informed referral decision, supported by an Explainable AI (XAI) evidence report, replacing the current practice of either ignoring cognitive concerns or initiating expensive specialist referrals based solely on subjective physician observation.

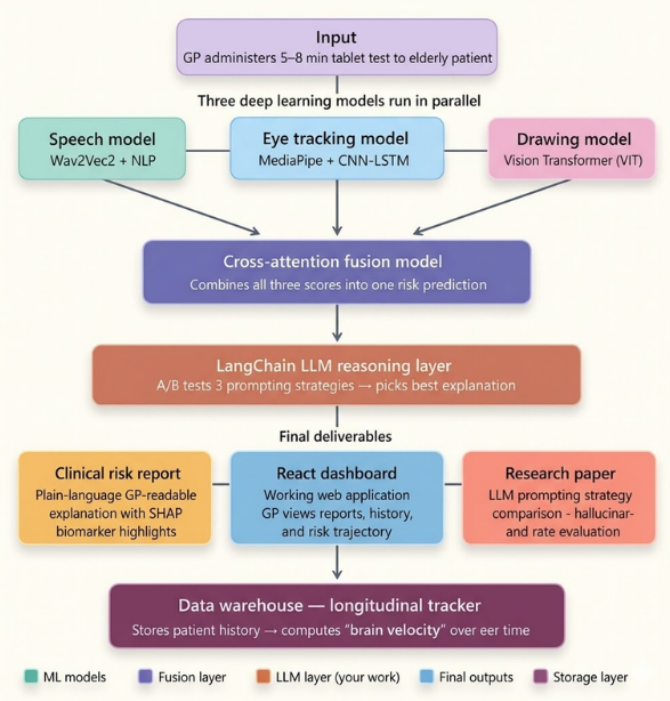


Fig. 1. Proposed architecture of the CogniSense framework.

The complete assessment protocol requires 5–8 minutes and is administered entirely through a standard Android tablet with a front-facing camera (minimum 8 MP, 30 fps) and microphone. No additional clinical equipment is required. The system captures three categories of digital biomarkers through three structured cognitive tasks, processes each through an independent deep learning pipeline, fuses the outputs via a Cross-Attention Transformer, and stores results in a longitudinal Data Warehouse for Brain Velocity computation.

B. Module 1: Speech and Acoustic Analysis

1) Task Protocol: The patient is administered the Cookie Theft Picture Description Task from the Boston Diagnostic Aphasia Examination — a standardized stimulus image for which the patient provides a 60-second verbal description recorded via the tablet microphone. This task is selected for its alignment with the DementiaBank Pitt Corpus training dataset [19], ensuring consistent stimulus-response conditions.

2) Feature Extraction Pipeline: Two parallel feature streams are extracted from the audio recording:

Acoustic Features (Language-Agnostic):

- Pause frequency and mean inter-word gap duration, extracted via WebRTC Voice Activity Detection (VAD).
- Fundamental frequency (F0): mean pitch, jitter, shimmer, and Harmonic-to-Noise Ratio (HNR).
- Mel-Frequency Cepstral Coefficients (MFCCs): 13-dimensional frame-level feature vector.
- Speech rate (syllables per minute) and vocal energy envelope.

Lexico-Semantic Features:

- Automatic transcription via Wav2Vec 2.0 XLS-R [11], supporting 128 languages.
- Type-Token Ratio (TTR) for lexical diversity.
- Propositional density and semantic coherence scoring via BERT sentence embeddings.
- Syntactic complexity via dependency parse tree depth (spaCy).

3) Model: A Bidirectional LSTM (BiLSTM) with attention mechanism processes the concatenated feature sequence to generate a 128-dimensional speech risk embedding. The model is trained and validated on the DementiaBank Pitt Corpus (167 dementia-diagnosed patients, 97 healthy controls) with target AUC-ROC > 0.82 against established baselines [4].

4) Language-Agnostic Design: By prioritizing acoustic over lexical features in the primary classification pathway, CogniSense maintains functionality across Hindi, Marathi, Tamil, Bengali, and other Indian regional languages. Wav2Vec 2.0 XLS-R's multilingual support enables transcript-based lexico-semantic features for major Indian languages as a secondary pathway.

C. Module 2: Oculomotor Analysis

1) Task Protocol: Three sequential oculomotor paradigms are administered:

Task A — Prosaccade (2 minutes): A stimulus dot appears randomly on the left or right of the screen. The patient must fixate upon the dot as rapidly as possible. This establishes baseline saccadic velocity and reaction time profiles.

Task B — Antisaccade (3 minutes, Primary Biomarker): A stimulus dot appears on one side; the patient must voluntarily fixate on the mirror-opposite side. This task directly engages the prefrontal cortex — specifically the frontal eye fields and anterior cingulate cortex — which are among the earliest neuroanatomical regions affected by tau pathology in AD [8]. The antisaccade error rate serves as the primary oculomotor biomarker.

Task C — Smooth Pursuit (1 minute): A dot traverses a sinusoidal horizontal path. The patient tracks it with their gaze. Pursuit gain and catch-up saccade frequency are computed.

2) Hardware-Free Implementation: Eye tracking is implemented entirely via MediaPipe Face Mesh [17], which maps 468 facial landmarks including 4 iris-specific landmark points per eye (indices 468–477) from standard front-facing camera video at 30 fps. Gaze vectors are computed from the iris centroid position relative to eye corner landmarks. Saccade detection employs a velocity threshold algorithm with onset defined at angular velocity $> 30°$/second.

Key metrics extracted include: antisaccade error rate (%), saccadic reaction time (ms), corrective saccade latency (ms), fixation stability (root mean square deviation of gaze coordinates), and smooth pursuit gain ratio.

3) Model: A CNN + LSTM architecture processes the gaze coordinate time-series to generate a 128-dimensional oculomotor risk embedding. Target correlation coefficient $r > 0.75$ with clinical ground truth antisaccade measures.

D. Module 3: Visuospatial Analysis — Digital Clock Drawing Test

1) Task Protocol: The patient draws a clock face on the tablet touchscreen (finger or stylus input) indicating the time "10 minutes past 11." This instruction is identical to the standard CDT administration protocol, enabling comparison against existing normative datasets.

2) Dual-Stream Feature Extraction: Static Features (Final Image Analysis): A Vision Transformer (ViT-Base/16) [13], pretrained on ImageNet and fine-tuned on CDT image datasets, extracts spatial features from the completed clock image: clock face completeness, numeral placement accuracy, hand length ratio, and hand angle accuracy for the target time.

Dynamic / Process Features (Temporal Sequence): These features are captured only in a digitized context and are invisible in the final static image:

- Stroke velocity profile across the drawing duration.
- Pen pressure variation (tremor quantification, stylus-enabled devices).
- Drawing order sequence — numerical sequence errors indicate planning dysfunction.
- Pause duration events mid-drawing.
- Total task completion time.
- Revision frequency (number of erasures or overdrawn strokes).

3) Model: The ViT output (spatial risk component) and an LSTM processing the stroke coordinate time-series (temporal risk component) are combined to produce a 128-dimensional CDT risk embedding. Target classification accuracy $> 85\%$ on the ADNI-CDT validation subset.

E. Module 4: Cross-Attention Transformer Fusion

The three 128-dimensional modality-specific risk embeddings — Speech (E_s), Oculomotor (E_o), and CDT (E_c) — are input to a Cross-Attention Transformer fusion module.

In contrast to late fusion approaches that compute a weighted average of independent risk scores, Cross-Attention Transformer fusion allows each modality to attend to the feature representations of the other modalities. This enables the model to learn clinically meaningful inter-modal correlations — for example, that the co-occurrence of elevated pause frequency (Speech) and high antisaccade error rate (Oculomotor) represents a multiplicatively higher risk signal than either biomarker in isolation.

The fusion module employs 2 transformer encoder layers with 4 attention heads. Monte Carlo Dropout is applied during inference to generate uncertainty estimates, enabling the system to report confidence intervals alongside risk classifications. The final output is a four-class risk classification:

- **Class 0:** Cognitively Normal
- **Class 1:** Subjective Cognitive Decline
- **Class 2:** Mild Cognitive Impairment (MCI)
- **Class 3:** High Risk — Urgent Referral

Target Fusion Model AUC-ROC > 0.88, Sensitivity $> 80\%$.

F. Module 5: Data Warehouse and Brain Velocity Tracking

Assessment results are stored in a Star Schema Data Warehouse with a central Fact Table (assessment_id, patient_id, date, speech_score, oculomotor_score, CDT_score, fused_score, risk_class, confidence) linked to Dimension Tables for Patient, Physician, and Time.

Brain Velocity is computed as the linear regression slope of fused cognitive scores over time (units: score points per month) for each patient with a minimum of two longitudinal assessment records. An alert threshold of -2.5 points per month triggers automatic risk class escalation regardless of absolute score. An LSTM-based forecasting sub-model projects the patient's future cognitive trajectory, estimating the time horizon to MCI threshold crossing.

Longitudinal validation employs publicly available datasets ADNI [20] and NACC, with Brain Velocity stability assessed via intra-class correlation coefficients (ICC). Target MAE < 0.05 per 6-month interval.

G. Module 6: Explainable AI Clinical Report

SHAP (SHapley Additive exPlanations) values [15] and transformer attention maps are computed for each assessment to generate a one-page Cognitive Risk Report readable by a General Physician without neurology training. The report includes: overall risk classification with confidence interval, modality-level scores with plain-language key findings, Brain Velocity trend visualization, projected cognitive trajectory, and a specific recommended clinical action formatted for direct attachment to specialist referral letters.

IV. Implementation Architecture

Datasets:

- *Speech:* DementiaBank Pitt Corpus [19] — 264 participants, 167 dementia-diagnosed.
- *Oculomotor:* GazeCapture (MIT) — 2.4M frames, 1,474 participants.
- *CDT Visuospatial:* ADNI [20], NHATS.
- *Longitudinal Validation:* ADNI [20], NACC (Uniform Data Set) [19].

V. Evaluation Framework

A. Module-Level Metrics

B. Clinical Performance Metrics

The system will be evaluated against standard clinical screening benchmarks:

- **Sensitivity (Recall):** Target $\geq 80\%$, prioritizing minimization of false negatives in a pre-clinical screening context.
- **Specificity:** Target $\geq 75\%$ to limit unnecessary specialist referrals.
- **Positive Predictive Value (PPV):** Evaluated against the current GP-only subjective assessment baseline.
- **AUC-ROC:** Primary scalar metric for discriminative performance across classification thresholds.

TABLE I
COGNISENSE TECHNOLOGY STACK

Layer	Component	Tool / Framework
Hardware	Primary interface	Android Tablet / Smartphone
Hardware	Oculomotor capture	Front-facing camera (8 MP+, 30 fps)
Hardware	Speech capture	Standard device microphone
Vision	Face mesh & iris	MediaPipe Face Mesh [17]
Speech	Feature extraction	Librosa, Parselmouth-Praat
Speech	Transcription & encoding	Wav2Vec 2.0 XLS-R [11]
NLP	Semantic analysis	spaCy, HuggingFace Transformers
DL	Speech model	BiLSTM + Attention
DL	Oculomotor model	CNN + LSTM
DL	CDT spatial model	ViT-Base/16 (PyTorch) [13]
DL	CDT temporal model	LSTM
Fusion	Multimodal integration	Cross-Attention Transformer
XAI	Feature attribution	SHAP [15], Attention Maps
Backend	API serving	FastAPI
DWH	Storage	PostgreSQL + Star Schema
MLOps	Experiment management	MLflow

TABLE II
PROPOSED EVALUATION METRICS AND TARGET BENCHMARKS

Module	Primary Metric	Target Benchmark
Speech & Acoustic	AUC-ROC, F1-Score	AUC > 0.82
Oculomotor	Antisaccade Correlation	$r > 0.75$
Visuospatial (CDT)	Classification Accuracy	$> 85\%$ on ADNI-CDT
Fusion Model	AUC-ROC, Sensitivity	AUC > 0.88, Sens. $> 80\%$
Brain Velocity	MAE	< 0.05 per 6-month interval

C. Explainability and Usability Evaluation

- **SHAP Validity:** Clinical expert review of attention maps against neurologically established early AD biomarkers.
- **Physician Usability:** System Usability Scale (SUS) administered to General Physicians in a simulated pilot setting.
- **Report Clarity Score:** Structured feedback from GP target users on Cognitive Risk Report comprehensibility.
- **Longitudinal Validation:** Brain Velocity stability measured via intra-class correlation coefficients (ICC) across simulated repeated assessments using ADNI and NACC datasets.

VI. DISCUSSION

A. Novelty and Differentiation

CogniSense addresses five simultaneous gaps unresolved by any existing system: multimodal biomarker fusion, smartphone hardware independence, language-agnostic design, longitudinal DWH-based trajectory tracking, and integration into the General Physician workflow. Existing solutions address at most one or two of these requirements in isolation [4]–[7], [9], [10]. The dynamic CDT feature extraction — capturing stroke velocity, drawing order, and pause duration from the digital drawing process — represents a contribution beyond any static image analysis approach in prior literature. The concept of Brain Velocity as a derived longitudinal metric from repeated low-cost assessments provides a temporal dimensionality to cognitive screening that eliminates the fundamental limitation of single-visit snapshot evaluations.

B. Cost and Accessibility Impact

CogniSense reduces the per-assessment cost from INR 15,000–80,000 (gold-standard diagnostics) to an estimated INR 5–15 (cloud compute cost per assessment), representing a cost reduction exceeding 2,500-fold. The zero-hardware-overhead deployment model eliminates capital expenditure barriers at Primary Health Centres (PHCs) and rural clinics. Assessment time is reduced from 2–4 hours (hospital visit) to 5–8 minutes, enabling cognitive screening integration into a standard GP consultation.

C. Language Inclusivity

India's linguistic diversity — 22 scheduled languages and hundreds of regional dialects — represents a critical barrier for English-centric cognitive screening tools. CogniSense's primary reliance on acoustic rather than lexical features ensures that pause patterns, pitch variation, speech rate, and vocal energy — clinically validated AD biomarkers [5] — remain extractable and interpretable regardless of the patient's language. The XLS-R multilingual Wav2Vec 2.0 variant extends transcript-based feature extraction to major Indian languages, providing a scalable path toward full linguistic coverage.

D. Ethical Considerations and Limitations

Patient data collected by CogniSense will be anonymized and encrypted at rest and in transit. The system's role as a triage tool — generating referral recommendations rather than diagnoses — mitigates risks associated with false positives, as all flagged cases require subsequent specialist confirmation. Current limitations include: the absence of prospective clinical validation data, dependency on camera quality and lighting conditions for oculomotor accuracy, and the requirement for at minimum two longitudinal data points before Brain Velocity computation becomes clinically meaningful. Future work will address these through a planned pilot deployment in collaboration with primary healthcare partner institutions.

VII. Conclusion

This paper has proposed CogniSense, a multimodal deep learning framework for pre-clinical Alzheimer's Disease screening designed for deployment by General Physicians in low-resource healthcare settings. By integrating acoustic speech biomarkers via Wav2Vec 2.0, oculomotor saccadic metrics via MediaPipe-based smartphone camera analysis, and visuospatial drawing features via a dual-stream ViT-LSTM architecture — fused through a Cross-Attention Transformer — CogniSense addresses the fundamental limitations of existing single-modality, hardware-dependent, and snapshot-oriented approaches.

The introduction of longitudinal Brain Velocity tracking via a structured Data Warehouse transforms the framework from a static screening test into a continuous cognitive health monitoring system, enabling earlier detection of trajectorial cognitive decline than any single-point assessment permits. The zero-hardware-overhead, language-agnostic, and GP-integrated design directly addresses India's three-dimensional Alzheimer's diagnostic crisis: the diagnostic gap, the accessibility crisis, and the cost barrier.

If deployed at scale, CogniSense has the potential to substantially increase the proportion of India's estimated 8.8 million affected individuals identified at the pre-clinical or MCI stage — the critical therapeutic window where disease-modifying interventions remain clinically effective — reducing irreversible neuronal loss, caregiver burden, and downstream healthcare system costs. This work contributes toward UN Sustainable Development Goals SDG 3 (Good Health and Well-Being), SDG 9 (Industry, Innovation and Infrastructure), and SDG 10 (Reduced Inequalities).

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